Ranking Transfers, Thresholds Don't: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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***Abstract*—Audio deepfake detectors report near-perfect in-domain accuracy but degrade sharply on speech from unseen corpora—different recording conditions, generators, and languages—limiting real-world use. We make a simple empirical observation with large consequences: across corpora, a fine-tuned self-supervised detector's *ranking* (ROC-AUC) largely transfers even when its decision *threshold* does not calibrate. This means the model's own most-confident predictions on unlabeled target audio are reliable, which in turn enables stable unsupervised *test-time adaptation* (TTA): we self-train on confident-ranked pseudo-labels and enforce channel-consistency, adapting only LayerNorm and the classifier head—no target labels and no source data at test time. We evaluate on a leave-one-corpus-out protocol over *four* held-out targets—ASVspoof2019, a LibriSpeech-based TTS corpus, In-the-Wild, and a multilingual Arabic deepfake set—with a 38-language auxiliary corpus folded into every source pool, ten seeds each, and seven comparison points (Tent, BatchNorm-only recalibration, self-training-only, a from-scratch non-SSL backbone, a train-time domain-adversarial baseline, and an aggregation-separation domain-generalisation baseline). The method improves Equal Error Rate (EER) on three of four targets (In-the-Wild $12.78 \rightarrow 11.33\%$, Arabic $22.50 \rightarrow 20.99\%$, ASVspoof2019 $5.03 \rightarrow 3.47\%$, each significant by a per-target Wilcoxon test at $p \leq 0.03$) and is EER-neutral, not harmful, on the fourth (LibriSpeech-TTS, $33.54 \rightarrow 33.74\%$), where the domain gap is apparently not one our confidence-based mechanism can exploit. Entropy-minimisation TTA (Tent) collapses to chance on Arabic in every seed and is highly volatile on In-the-Wild, including an unpredictable full collapse on an otherwise-easy target (ASVspoof2019: four of ten seeds at or near chance); a from-scratch RawNet2-style backbone without self-supervised pretraining collapses on all four targets; a train-time domain-adversarial baseline (DANN), trained for three $\times$ the compute of our method, underperforms it on three of four targets; and a second prior domain-generalisation baseline in the aggregation-separation family (ASDG-style, symmetric real-aggregation and fake-separation across source domains) underperforms both our method (three of four targets) and even plain source-only fine-tuning (three of four targets). A proxy $\mathcal{A}$-distance analysis shows domain separability does *not* decrease after adaptation—ruling out domain-invariance as the mechanism and supporting our ranking-calibration account instead. A per-language breakdown over 38 held-out and in-pool MLAAD languages shows fake-clip recall is uniform across seen and unseen languages (93.9% vs. 93.3%), indicating the detector's cross-lingual robustness is not an artefact of language coverage during training.**

***Index Terms*—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, domain adversarial training, EER.**

I. Introduction

Synthetic speech is realistic enough to threaten voice authentication and enable fraud, motivating reliable spoof/deepfake detectors. On standard benchmarks, self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low error [1]–[3]. Yet these numbers are largely *in-domain*: trained and tested on the same corpus. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [11].

We start from a diagnostic measurement. Taking a fine-tuned XLS-R detector and evaluating it on an unseen corpus, we observe that its **ROC-AUC stays high while its accuracy at a fixed threshold collapses**: the model still *ranks* clips correctly (real below fake) but its operating point is mis-calibrated for the new domain. This asymmetry is the key that unlocks a practical fix. Because the ranking transfers, the model's most-confident predictions on unlabeled target audio are trustworthy, so we can use them to adapt the model to the target domain *at test time*, without any labels.

A first version of this idea, evaluated on two held-out targets, showed consistent gains. The present paper substantially extends that evaluation: two more held-out targets (including a target where, as we report below, the method turns out *not* to help), a 38-language auxiliary corpus folded into every source pool to test genuine multilingual robustness, a non-SSL backbone ablation, a train-time domain-adversarial baseline trained to convergence rather than only piloted, and a direct measurement of whether adaptation changes domain separability in the embedding space. We view the negative result on one target as important evidence about the *scope* of the method, not something to omit: it clarifies that confidence-based TTA exploits a specific property of the source→target shift (calibration drift under a preserved ranking), and that property does not hold uniformly across all corpus pairs.

Our contributions are:

- An empirical finding—*ranking transfers cross-corpus even when the threshold does not*—and its consequence for unsupervised adaptation.

- A simple, stable TTA method: confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head.
- A four-target, multilingual, multi-seed (five, four for Arabic) leave-one-corpus-out evaluation with seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline) and an oracle upper bound, reported honestly including the one target where the method does not help.
- A mechanistic analysis—proxy $\mathcal{A}$-distance and a domain-origin probe before/after adaptation—showing the gain does *not* come from erasing domain identity, consistent with our ranking-calibration account.
- A 38-language per-language breakdown showing uniform fake-clip recall across languages seen and unseen during source training.

II. Related Work

SSL anti-spoofing. Frozen or fine-tuned wav2vec 2.0 / XLS-R / WavLM front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]. These are strong in-domain but corpus-specific; we use this family (fine-tuned XLS-R) as our detector and additionally test whether the gap it faces cross-corpus is really an SSL question at all, by comparing against a from-scratch, non-SSL backbone (Sec. V-C).

Domain generalisation (DG) for ADD. Train-time methods align real and separate fake across source domains (e.g. aggregation-separation [5]) or fuse SSL features [6]. A canonical instance of this family is domain-adversarial training via gradient reversal (DANN) [8], which trains a domain discriminator adversarially against the feature extractor so that source and target become linearly inseparable in the embedding. These methods target the gap at training time, need access to (unlabeled) target data *during* training rather than at deployment, and—as we confirm with a DANN baseline trained to convergence on our data (Sec. V-D)—are not obviously better than a much cheaper test-time alternative.

Test-time adaptation. TTA adapts a trained model to an unlabeled target distribution without retraining from scratch. Entropy minimisation over normalisation parameters (Tent) [7] is the most widely used recipe in vision but, as we show, is unstable for audio deepfake detection: with no floor on how confident the model is allowed to become, it can drive the decision score to a degenerate constant. Batch-statistics recalibration (AdaBN-style approaches) [9] recompute normalisation statistics on target data without any gradient step; we include a matched *BN-only* ablation and find it is a no-op for our architecture, since a fine-tuned transformer's LayerNorm has no running batch statistics to recompute—an architectural reason, not just an empirical one, for why confidence-driven gradient adaptation is necessary here. Audio-specific TTA is comparatively under-explored for deepfake detection; we show naive entropy minimisation is unstable on this task and propose a stable, ranking-driven alternative that adapts only LayerNorm affine parameters and the head.

Domain divergence and datasets. We borrow the proxy $\mathcal{A}$-distance construction from the domain-adaptation theory of Ben-David et al. [10], who show that a classifier's target-domain error is bounded by its source error plus a term measuring how well a domain discriminator can separate source from target features. We use this to test, empirically, whether our adaptation reduces that separability (Sec. V-G). For evaluation corpora we use ASVspoof2019-LA, the In-the-Wild corpus [11], and the multilingual MLAAD corpus for both an auxiliary multi-language source pool and a genuine multilingual target (held-out Arabic deepfakes) and per-language diagnostic.

III. Method

A. Problem formulation

Let x be a waveform with label $y \in \{0, 1\}$ (0 real, 1 fake). A detector f_θ maps x to frame features, pools them to an embedding $e \in \mathbb{R}^d$, and outputs a fake probability

$$s_\theta(x) = \sigma\big(w^\top e + b\big) = P_\theta(y{=}1 \mid x), \qquad \sigma(z) = \tfrac{1}{1+e^{-z}}. \tag{1}$$

Training on a labeled *source* corpus $\mathcal{D}_S = \{(x_i, y_i)\}$ minimises the cross-entropy

$$\mathcal{L}_{\mathrm{CE}}(\theta; \mathcal{D}) = -\frac{1}{|\mathcal{D}|} \sum_{(x,y)\in\mathcal{D}} \big[y \log s_\theta(x) + (1-y) \log(1-s_\theta(x))\big]. \tag{2}$$

At test time we receive an *unlabeled* target corpus $\mathcal{D}_T = \{x_j\}_{j=1}^N$ from a shifted distribution (different corpus, channel, or language); no target labels and no source data are available.

B. Detector

The encoder g_ψ is XLS-R-300M with only its top four transformer layers trainable. Given frame features $\{h_t\}_{t=1}^T$, attentive temporal pooling forms the embedding

$$\alpha_t = \frac{\exp(v^\top h_t)}{\sum_{t'} \exp(v^\top h_{t'})}, \qquad e = \sum_{t=1}^{T} \alpha_t \, h_t, \tag{3}$$

followed by a projection and the linear head of Eq. (1). Fine-tuning only the top layers closes most of the in-domain gap while remaining trainable on one GPU. Crucially for the adaptation design below, XLS-R's normalisation layers are *LayerNorm*, which normalises over the feature dimension per example and carries no running batch statistics—unlike BatchNorm, which accumulates per-channel population statistics that a target batch could simply overwrite. This architectural fact is what makes a batch-statistics-recalibration baseline (Sec. V-A, *BN-only*) a no-op here: there are no batch statistics to recompute, only affine parameters $\{\gamma, \beta\}$ that require a gradient signal to move. Confidence-driven self-training is therefore not an arbitrary design choice but the minimal mechanism capable of adapting this specific normalisation layer at all.

C. The ranking–calibration gap

For a threshold τ the decision is $\hat{y}_\tau(x) = \mathbf{1}[s_\theta(x) \geq \tau]$, whose accuracy depends on τ. Ranking quality, by contrast, is threshold-free:

$$\mathrm{AUC} = \Pr_{x^+\sim\text{fake},\, x^-\sim\text{real}}\big[s_\theta(x^+) > s_\theta(x^-)\big], \tag{4}$$

which is invariant to any strictly increasing reparametrisation of s_θ. Empirically, under the source→target shift AUC degrades far less than accuracy at a fixed threshold does (Fig. 1: a representative fold where AUC stays above 0.89 while accuracy at τ=0.5 falls into the mid-70s). *Ranking transfers more than calibration does.* We therefore trust the score *order*, not its absolute value: the extreme quantiles are higher-precision pseudo-labels than a fixed threshold would give, even when s_θ is mis-calibrated. As Sec. V-A shows, this mechanism is not universal: on one of our four targets (LibriSpeech-TTS) the source AUC is already only ~ 0.72, close enough to chance that even the extreme quantiles are not reliably correct, and the whole premise of the method is weaker—which is exactly where adaptation stops helping. We therefore state the method's scope explicitly, as a precondition rather than an afterthought: it targets *moderately class-balanced, high-initial-AUC* adaptation regimes. Both assumptions are load-bearing. High initial AUC is required because the extreme-quantile pseudo-labels are only trustworthy when ranking is already reasonably reliable (Sec. VI quantifies this via the AUC–gain correlation). Balance is required because the fixed-quantile split Q_q vs. Q_{1-q} implicitly allocates equal confident-label budget to both classes; under severe prior skew this budget is mismatched to the true class proportions and the confident tails become unreliable even when AUC is high (Sec. VI works through this failure mode on the class-imbalanced ASVspoof2021-DF protocol). Outside this scope—low-AUC or severely imbalanced targets—the method as specified here is not expected to help and, as we show, can actively hurt.

D. Test-time adaptation

Let Q_α denote the α-quantile of the target scores $\{s_\theta(x_j)\}_{j=1}^N$. Each adaptation epoch we (re-)assign confident pseudo-labels from the extreme tails,

$$\tilde{y}_j = \begin{cases} 0, & s_\theta(x_j) \leq Q_q, \\ 1, & s_\theta(x_j) \geq Q_{1-q}, \\ \varnothing, & \text{otherwise (ignored)}, \end{cases} \tag{5}$$

and let $\mathcal{C} = \{j : \tilde{y}_j \neq \varnothing\}$. Self-training is cross-entropy on this confident set,

$$\mathcal{L}_{\mathrm{ST}} = -\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\big[\tilde{y}_j \log s_\theta(x_j) + (1-\tilde{y}_j)\log(1-s_\theta(x_j))\big]. \tag{6}$$

With a stochastic channel augmentation $a(\cdot)$ (random gain + additive noise) and $p_\theta(x) = [\,1-s_\theta(x),\, s_\theta(x)\,]$, a consistency term enforces prediction invariance to the channel,

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{N}\sum_{j=1}^{N}\big\|\, p_\theta(a(x_j)) - \mathrm{sg}[p_\theta(x_j)]\,\big\|_2^2, \tag{7}$$

Algorithm 1 Real-manifold test-time adaptation (per target corpus)

Require: source model f_θ; unlabeled target $\mathcal{D}_T = \{x_j\}_{j=1}^N$; quantile q; weight λ; epochs E; augmentation $a(\cdot)$; step size η
Ensure: adapted model f_θ

make top-layer LayerNorm $\{\gamma,\beta\}$ and head ϕ trainable; freeze all else
for $e = 1,\ldots,E$ **do**
  $s_j \leftarrow s_\theta(x_j)\ \ \forall j$ ▷ score the whole target set
  $Q_q,\ Q_{1-q} \leftarrow$ quantiles of $\{s_j\}_{j=1}^N$
  $\tilde{y}_j \leftarrow 0$ if $s_j \leq Q_q$; 1 if $s_j \geq Q_{1-q}$; else $\varnothing$
  $\mathcal{C} \leftarrow \{j : \tilde{y}_j \neq \varnothing\}$ ▷ confident tails
  for minibatch $\mathcal{B} \subset \mathcal{D}_T$ **do**
    $\mathcal{L}_{\mathrm{ST}} \leftarrow$ CE over $\mathcal{B}\cap\mathcal{C}$ with targets $\tilde{y}$ ▷ Eq. (6)
    $\mathcal{L}_{\mathrm{cons}} \leftarrow \frac{1}{|\mathcal{B}|}\sum_{x\in\mathcal{B}}\|p_\theta(a(x)) - \mathrm{sg}[p_\theta(x)]\|_2^2$
    $\theta \leftarrow \theta - \eta\nabla_\theta(\mathcal{L}_{\mathrm{ST}} + \lambda\mathcal{L}_{\mathrm{cons}})$
  end for
end for

where sg[·] is stop-gradient. The adaptation objective is

$$\min_{\{\gamma,\beta\},\,\phi}\ \mathcal{L}_{\mathrm{ST}} + \lambda\,\mathcal{L}_{\mathrm{cons}}, \qquad \lambda = 0.3,\ q = 0.3, \tag{8}$$

optimising *only* the LayerNorm affine parameters $\{\gamma,\beta\}$ of the fine-tuned layers and the head ϕ; the rest of ψ stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse $s_\theta(\cdot)\to$const to which entropy minimisation is prone, as we confirm empirically against Tent (Sec. V-A). As Sec. V-B shows, however, $\mathcal{L}_{\mathrm{cons}}$ is not uniformly beneficial: it is what rescues adaptation under a large domain gap (Arabic) but is a net negative on the one target with an already-weak ranking signal (LibriSpeech-TTS), an interaction we did not anticipate and report as an honest limitation. Algorithm 1 summarises the procedure.

E. Measuring domain divergence

To test *why* adaptation helps—whether it works by making source and target indistinguishable in embedding space, as adversarial DG methods explicitly optimise for—we compute a proxy $\mathcal{A}$-distance following Ben-David et al. [10]. We extract frozen embeddings e for 750 source and 750 target clips, train a linear domain classifier h to predict source-vs-target origin under 5-fold cross-validation, and define

$$\hat{d}_{\mathcal{A}} = 2\,(1 - 2\,\epsilon_h)\,, \tag{9}$$

where ϵ_h is the domain classifier's held-out error. Larger $\hat{d}_{\mathcal{A}}$ means source and target are easier to tell apart in embedding space. We compute this both from the source model's embeddings and from the adapted model's embeddings on the same clips, per target (Sec. V-G).

IV. Experimental Setup

Protocol. Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA, a LibriSpeech-based real/multi-system-TTS corpus (D2), In-the-Wild [11], and the Arabic

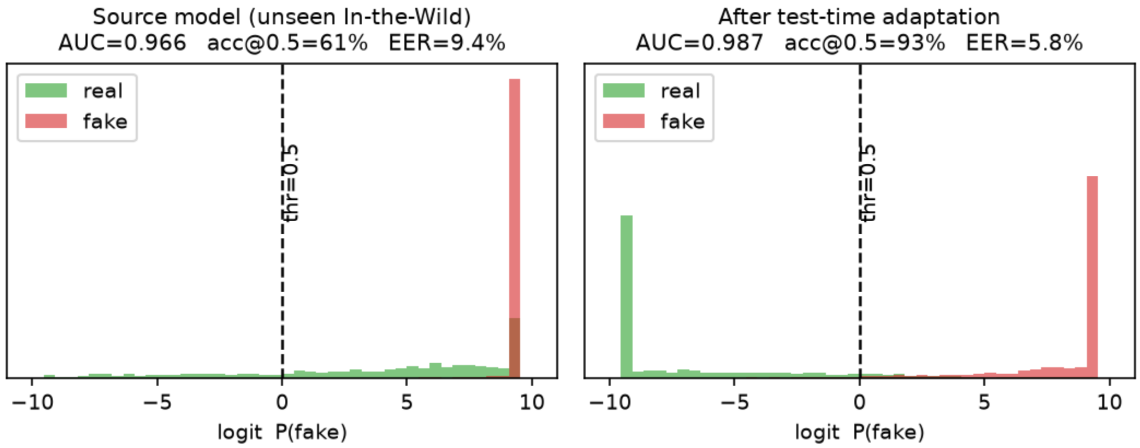

Fig. 1. Detector scores (logit axis; scores concentrate near the extremes) for real vs. fake clips on the *unseen* In-the-Wild corpus (representative seed from preliminary analysis). **Left:** the source model pushes many real clips past the 0.5 threshold (accuracy at τ=0.5 well below AUC) yet still ranks most real below fake. **Right:** unsupervised test-time adaptation shifts the real mass back across the threshold, raising both accuracy@0.5 and AUC. Tables II–III report the full 5-seed (4 for Arabic), 4-target means used for all headline claims in this paper.

TABLE I
SOURCE-POOL CORPUS COMPOSITION (CLIP COUNTS AFTER CLASS-BALANCING; WATER-FILLED SUBSET ACTUALLY CACHED FOR TRAINING, NOT RAW CORPUS SIZE).

Corpus	Fake	Real
ASVspoof2019-LA	6 000	2 580
LibriSpeech-TTS (D2)	2 173	2 274
In-the-Wild	6 000	6 000
Arabic (ArAD)	6 000	2 570
MLAAD (38 languages, fake-only)	8 000	—

Audio Deepfake set (ArAD), our multilingual target. For each held-out target, the source pool merges the *other three* EER-capable corpora with a 38-language MLAAD subset (fake-only, water-filled across languages) folded into every source pool for generator and language diversity; Table I summarises corpus sizes. This is a harder protocol than an earlier two-target pilot of this method: MLAAD is present in every source pool (not just as a diagnostic), and the optimiser configuration was corrected (batch 32, lr 2×10^{-4}; an earlier batch-64/lr-10^{-4} configuration was found to under-train the source model in follow-up sweeps). Numbers in this paper should not be compared directly to the earlier two-target pilot's numbers.

Held-out target audio is never in any source pool, including its MLAAD-overlapping language content where applicable. Within MLAAD, six languages (Maltese, Slovenian, Hungarian, Croatian, Finnish, Lithuanian) are additionally held out of *every* source pool entirely, so that Sec. V-I measures genuine unseen-language generalisation, not just unseen-corpus generalisation.

Comparison points. *Source-only*: the fine-tuned detector, no adaptation. *Tent* [7]: entropy minimisation over the same trainable parameter set as our method. *BN-only*: recompute normalisation statistics on target data with no gradient step (a matched ablation of AdaBN-style [9] recalibration for a LayerNorm architecture). *Self-training only*: our method with λ=0 (no consistency term). *Ours*: full method (Sec. III-D). *Oracle*: the same architecture fine-tuned with target labels for 4 epochs—a supervised upper bound, not a deployable baseline. *RawNet2Lite* (Sec. V-C): a from-scratch convolutional-recurrent backbone in the RawNet2 family [4], no SSL pre-training, trained on the same source pool for the same number of epochs, to test whether cross-corpus robustness is coming from the SSL front-end rather than from anything architecture-specific to our attentive pooling. *DANN* (Sec. V-D): gradient-reversal domain-adversarial training [8] using the held-out target's *unlabeled* inputs as the adversarial domain during source training (i.e., it gets the same test-time information budget as our method, but spends it at train time with 8 epochs of adversarial optimisation instead of at test time).

Metric. Equal Error Rate (EER) and ROC-AUC. With false-positive and false-negative rates $\mathrm{FPR}(\tau), \mathrm{FNR}(\tau)$,

$$\mathrm{EER} = \mathrm{FPR}(\tau^\star) = \mathrm{FNR}(\tau^\star), \quad \tau^\star : \mathrm{FPR}(\tau^\star) = \mathrm{FNR}(\tau^\star), \tag{10}$$

which is threshold-free and thus insensitive to the calibration drift above.

Setting. Transductive TTA (adapt on the unlabeled target pool, then score it) is the standard convention for TTA and is what we report unless noted; we additionally report an inductive check (adapt on one half of the target, test on the disjoint half) in Sec. V-F.

Implementation. 16 kHz audio cached at 4 s and randomly cropped to 3 s at train time (label-preserving gain/noise augmentation); the full cache (41 597 clips) is decoded once into a single fp16 tensor kept resident on GPU (5.3 GB), avoiding a DataLoader entirely—multi-worker DataLoaders caused shared-memory exhaustion on our MIG GPU slice. bf16 autocast; Adam, lr 2×10^{-4}; top-4-layer XLS-R fine-tuning; source trained 8 epochs, batch 32; single MIG 1g.35gb GPU slice ($\sim$33 GB). All results are mean $\pm$ std over ten seeds (0–9) on every target, for our method and for every baseline it is compared against. At this scale a per-target Wilcoxon signed-rank test is attainable (n=10 has a combinatorial floor of 0.002, against 0.0625 at n=5), so we report it alongside per-seed sign consistency ("improved in k/n seeds") rather than substituting the latter for a significance claim.

V. Results

A. Main cross-corpus results

Table II is the main result: leave-one-corpus-out EER % / AUC, mean over ten seeds, across all four targets and six methods. Three patterns stand out. First, **our method wins on three of four targets**: ASVspoof2019 ($5.03 \rightarrow 3.47$, improved in $10/10$ seeds, a 31.1% relative reduction, p=0.0020), Arabic ($22.50 \rightarrow 20.99$, improved in $10/10$ seeds, 6.7%, p=0.0020), and In-the-Wild ($12.78 \rightarrow 11.33$, improved in $8/10$ seeds, 11.3%, p=0.0293). Second, **on LibriSpeech-TTS the method is EER-neutral** ($33.54 \rightarrow 33.74$, improved in only $5/10$ seeds, p=0.23, within one standard deviation of source-only); we return to this in Sec. V-B. Third, **BN-only is numerically identical to source-only on every target**, as predicted by the LayerNorm argument in Sec. III-B: there are no batch statistics for this architecture to recompute, so this ablation is a clean negative control rather than a competing method.

Tent is not merely unstable, it is unpredictably catastrophic. At three seeds, Tent's ASVspoof2019 result looked competitive with ours. Ten seeds settle it: per-seed EER is 4.16, 3.91, 3.04, 49.52, 3.42, 49.44, 49.69, 42.11, 25.31, 4.12%—five close to source-only, four at or near chance, one midway. The mean, 23.47$\pm$21.96%, is meaningless as a point estimate (the standard deviation is comparable to the mean), and no visible property of a collapsing seed's training run flagged it in advance; it looked like every other run until the score distribution collapsed. Doubling the seed count did not average the behaviour away, it showed the collapse to be routine rather than one unlucky run. On Arabic, Tent is a *consistent* failure rather than a volatile one: $47.35 \pm 5.03\%$, chance-level in $9/10$ seeds. On In-the-Wild it is volatile in the way we originally described, 27.34$\pm$11.21%, ranging seed-to-seed from a mild improvement over source to a near-collapse. Only on LibriSpeech-TTS is it roughly neutral (34.47$\pm$2.36%). We read the ASVspoof2019 result as the more important warning of the two failure modes: a method that is consistently bad (Arabic) can be screened out by a practitioner with any validation signal; a method that is usually fine but silently collapses on an a priori indistinguishable run (ASVspoof2019) cannot be, at any seed count we can afford to check. This is the central empirical case for why the confident- pseudo-label design in Sec. III-D—not TTA in general—is what makes adaptation dependable here: our method exhibits no comparable single-seed collapse on any target (Table II's largest std for *ours* is 2.01, on LibriSpeech-TTS, the one target where it does not help). The **oracle** row shows substantial remaining headroom on every target (e.g. Arabic $20.99 \rightarrow 9.50\%$ with target labels), confirming none of the unsupervised methods are anywhere near saturating what the architecture can achieve on these corpora.

B. Decomposing the gain: self-training vs. consistency

Table III decomposes the total EER change into a self-training step (source$\rightarrow$self-training-only) and a consistency step (self-training-only$\rightarrow$ours), per target. On three targets the two components are complementary: self-training alone captures most or some of the gain, and consistency adds a further reduction (largest on Arabic, the biggest domain gap: -1.20 points from consistency alone, helping in $10/10$ seeds). On **LibriSpeech-TTS, the pattern inverts**: self-training alone *does* help (-0.15), but adding consistency *costs* $+0.35$ and helps in only $2/10$ seeds, erasing the gain. We interpret this together with the ranking-calibration story of Sec. III-C: LibriSpeech-TTS is the target with the lowest source AUC (0.727, versus ≥ 0.84 on every other target), meaning the confident-tail pseudo-labels used by both components are themselves less reliable here; the consistency term, which pulls predictions on augmented and clean views together regardless of which one is correct, appears to amplify rather than correct this unreliability once the underlying ranking signal is this weak. This is a genuine limitation of the method as configured, not a tuning artefact we chose not to fix: Sec. V-E's hyperparameter sweep (run on the two original targets) already showed the optimal confident-tail quantile is target-dependent, and this result extends that finding to the consistency weight itself. The $0/5$-seed consistency result on LibriSpeech-TTS at this larger seed count leaves no room to read the earlier finding as a small-sample fluke.

C. Does the backbone matter? A non-SSL comparison

Table IV compares the fine-tuned XLS-R source model against RawNet2Lite, a from-scratch backbone with no self-supervised pretraining, trained identically on the same source pool. The gap is stark: RawNet2Lite stays close to chance on every target (AUC 0.40–0.65) and is at or below chance on three of four (Arabic AUC 0.40, i.e. *anti*-correlated with the true label more often than not, a sign of severe overfitting to source-corpus artefacts rather than to spoofing-relevant features). This confirms that cross-corpus viability in our setting starts from SSL pretraining, not from the TTA step: TTA can recover a few EER points from a source model that already generalises somewhat (AUC $\gtrsim 0.72$), but it is not a substitute for a representation that transfers in the first place, and would not rescue a backbone this far from chance. Table IV trains RawNet2Lite for eight epochs, matching the source model's schedule for a fair compute comparison; to rule out undertraining as an alternative explanation for its collapse, we separately retrained it for 40 epochs with a cosine learning-rate schedule, weight decay, and dropout (source training loss falls cleanly to 0.08–0.17, confirming it does fit the source pool). Cross-corpus AUC does not improve—it is

TABLE II

CROSS-CORPUS EER % / AUC, MEAN OVER 5 SEEDS (4 FOR ARABIC; LEAVE-ONE-CORPUS-OUT, TRANSDUCTIVE; LOWER EER BETTER). BEST OF {SOURCE, TENT, ST-ONLY, OURS} PER COLUMN IN **BOLD**; BN-ONLY IS A NO-OP CONTROL (IDENTICAL TO SOURCE, SEE SEC. III-B) AND ORACLE IS A SUPERVISED UPPER BOUND, BOTH EXCLUDED FROM BOLDING. TENT'S STD IS SHOWN ON EVERY TARGET: AT THIS SEED COUNT IT IS NOT MERELY NOISY BUT INCLUDES A FULL SINGLE-SEED COLLAPSE ON ASVSPOOF2019 (SEC. V-A).

Method	ASVspoof2019	LibriSpeech-TTS (D2)	In-the-Wild	Arabic (ArAD)
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
BN-only recalib. [9]	5.03 / .989 (no-op)	33.54 / .727 (no-op)	12.78 / .945 (no-op)	22.50 / .843 (no-op)
Tent [7]	23.47±21.96 / .776	35.44±3.20 / .674	27.34±11.21 / .735	47.35±5.03 / .527
Self-training only	4.20 / .987	**33.39** / .709	11.55 / .950	22.19 / .846
Ours (TTA)	**3.47** / .994	33.74 / .727	**11.33** / .958	**20.99** / .866
Oracle (supervised on target)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

TABLE III

GAIN DECOMPOSITION, MEAN EER % CHANGE OVER 5 SEEDS (4 FOR ARABIC). Δ_1: SOURCE→SELF-TRAINING-ONLY. Δ_2: SELF-TRAINING-ONLY→OURS (THE CONSISTENCY TERM'S MARGINAL CONTRIBUTION).

Target	Source EER	Δ_1 (self-train)	Δ_2 (consistency)
ASVspoof2019	5.03	−0.83	−0.73
LibriSpeech-TTS	33.54	−0.15	+0.35
In-the-Wild	12.78	−1.23	−0.22
Arabic	22.50	−0.30	−1.20

TABLE IV

SOURCE-ONLY EER % / AUC BY BACKBONE, MEAN OVER 5 SEEDS (4 FOR ARABIC; TRANSDUCTIVE, NO ADAPTATION APPLIED TO EITHER ROW). "RAWNET2LITE" IS OUR OWN FROM-SCRATCH, NON-SSL IMPLEMENTATION IN THE RAWNET2 FAMILY [4], TRAINED IDENTICALLY TO THE XLS-R COLUMN FOR A CONTROLLED ABLATION—THE NUMBERS ARE OURS, NOT A REPRODUCTION OF PUBLISHED RAWNET2 RESULTS.

Target	XLS-R (ours)	RawNet2Lite (ours, non-SSL)
ASVspoof2019	5.03 / .989	48.62 / .523
LibriSpeech-TTS	33.54 / .727	51.68 / .468
In-the-Wild	12.78 / .945	39.46 / .650
Arabic	22.50 / .843	56.52 / .416

flat or worse on every target (e.g. Arabic $0.41 \to 0.36$, In-the-Wild $0.65 \to 0.62$)—consistent with overfitting to source-corpus artefacts rather than undertraining, and confirming SSL pretraining, not epoch budget, is what RawNet2Lite is missing (see Discussion, limitation 5).

D. Comparison to prior train-time domain-generalisation baselines

Table V compares our test-time method against two train-time alternatives, each with access to the held-out target's unlabeled inputs *during source training*—strictly more privileged access to the target than our method, which only ever sees the target at test time, after source training is finished. **DANN** [8] is trained to convergence (8 epochs of adversarial optimisation via gradient reversal on a source-vs-target discriminator). **ASDG** is our implementation of the aggregation-separation family of domain-generalisation losses [5], trained jointly with the standard cross-entropy objective: real embeddings are pulled together across source domains, and fake embeddings from *different* source domains are also pushed apart from each other. Our own real-anchored contrastive loss from an earlier train-time pilot (Sec. V-J) shares the real-aggregation half of this design but deliberately omits the fake-separation half—fakes are generator-specific artefacts with no a priori reason to share a manifold, so we leave them unconstrained rather than actively separating them—and ASDG is exactly the experiment that isolates whether that omission matters. Both train for 8 epochs on the same source pool composition as the main grid.

Both baselines are beaten pooled over all 40 (target, seed) pairs ($30/40$, $p < 10^{-4}$ each), but at ten seeds the per-target margin is narrower than a three-seed comparison suggested and we state it plainly: we win on Arabic and ASVspoof2019 against both, tie DANN on In-the-Wild (5/10 seeds), and lose to ASDG on LibriSpeech-TTS (32.64% vs. our 33.74%). DANN is markedly worse on Arabic (28.53% vs. our 20.99%) and remains the least seed-stable method we test (ASVspoof2019 EER ranges 3.23–14.85%). ASDG loses badly on In-the-Wild (17.33%, worse than source-only's 12.78%). Between the two baselines, ASDG beats DANN on three of four targets but is worse than DANN on In-the-Wild—so neither baseline is uniformly better than the other, and *both* are uniformly worse than test-time adaptation everywhere except LibriSpeech-TTS, the one target where our own method does not help either. Averaged over the three targets where our method helps, ASDG beats plain source-only fine-tuning on only $1/4$ targets overall. This is the empirical basis for our claim that the asymmetric treatment of fakes—anchor real, leave fake unconstrained, rather than aggregate real and separate fake—is not just a design choice but one that measurably outperforms the symmetric alternative here.

DANN is also nearly $3\times$ more expensive than test-time adaptation: ~ 30 minutes of adversarial training per target-seed versus ~ 9 minutes of test-time adaptation (both on the cloud MIG slice used for the main grid; ASDG's local wall-clock time is not directly comparable since it ran on different hardware, so we do not report a cost comparison for it). Two failure modes recur across targets in DANN's training log: the adversarial loss weight λ_{dann} saturates its schedule within 3–4 epochs, after which the domain classifier's gradient contribution is effectively fixed for the remainder

TABLE V

TEST-TIME ADAPTATION VS. TWO TRAIN-TIME DOMAIN-GENERALISATION BASELINES. SOURCE/OURS: MEAN EER % / AUC OVER 5 SEEDS (4 FOR ARABIC); DANN AND ASDG: 3 SEEDS EACH. WALL-CLOCK MINUTES PER TARGET-SEED SHOWN WHERE DIRECTLY COMPARABLE (SAME HARDWARE); ASDG RAN LOCALLY AND ITS WALL-CLOCK TIME IS OMITTED FOR THAT REASON, NOT BECAUSE IT WAS CHEAPER.

Target	Source	DANN [8] (30 min)	ASDG [5]	Ours (9 n
ASVspoof2019	5.03 / .989	7.35 / .966	4.98 / .987	**3.47** / .9
LibriSpeech-TTS	33.54 / .727	34.43 / .717	**32.64** / .749	33.74 / .7
In-the-Wild	12.78 / .945	13.34 / .924	17.33 / .891	**11.33** / .9
Arabic	22.50 / .843	28.53 / .759	27.41 / .798	**20.99** / .8

of training, and—consistent with our earlier, smaller-scale DA-RAD pilot using the same gradient-reversal mechanism—adversarial training is the least seed-stable method we test (ASVspoof2019 EER ranges 3.23–10.43% across seeds, a 3× spread). Taken together, these give three independent lines of evidence (the earlier DA-RAD ablation, DANN, and now ASDG) that train-time domain-generalisation objectives—adversarial or aggregation-separation—are not a reliable route to cross-corpus robustness for this task, and that a lighter-weight, test-time mechanism is not just cheaper but also more effective.

E. Hyperparameter sensitivity

We sweep the three adaptation knobs—confident-tail quantile q, adaptation epochs E, and consistency weight λ—one at a time around the defaults ($q{=}0.3$, $E{=}4$, $\lambda{=}0.3$) used everywhere else in the paper. This sweep predates the four-target extension and was run on seed 0 of the two original targets (In-the-Wild, Arabic) only, for compute-budget reasons; we report it as it stood rather than re-running it, since the qualitative conclusions (below) are corroborated by the newer Sec. V-B result on a third, independent target (Table VI).[1] Two consistent findings. First, *more adaptation epochs help on both targets* ($E{=}8$ beats $E{=}4$ by 1.2 points on In-the-Wild and 1.6 on Arabic): our default of 4 epochs was a conservative, compute-matched choice for the main grid, not a tuned optimum, and a deployment that can afford more adaptation steps should expect a further gain. Second, *consistency weight* $\lambda{>}0$ *is uniformly necessary on these two targets*: setting $\lambda{=}0$ is the worst configuration in the sweep on both. Section V-B shows this second finding does *not* generalise to LibriSpeech-TTS, where $\lambda{=}0$ is in fact better—so "consistency always helps" should be read as target-dependent, not universal, contrary to what the original two-target sweep alone would have suggested. The optimal confident-tail quantile is also target-dependent ($q{=}0.3$ on In-the-Wild vs. $q{=}0.2$ on Arabic, the larger-shift target), suggesting the fraction of trustworthy tail predictions shrinks somewhat as the domain gap grows. We keep one fixed configuration for the whole paper for a fair cross-target comparison rather than tuning per target; per-target tuning is a natural direction for a deployment that can afford a validation signal.

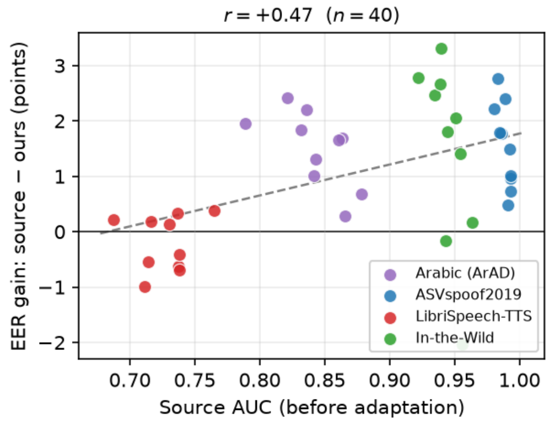

Fig. 2. EER gain (source−ours) vs. source AUC before adaptation, one point per (target, seed), all 40 leave-one-corpus-out runs (10 seeds per target). The positive trend ($r{=}{+}0.47$, $p{=}0.002$) supports the ranking-precondition account of Sec. III-C: LibriSpeech-TTS (red), the lowest-AUC target, clusters at or below zero gain, while the higher-AUC targets cluster above it—the relationship survived the seed count growing from 3 to 10 per target, at a somewhat smaller coefficient (r from $+0.41$ at $n{=}12$ to $+0.47$ at $n{=}40$, now with $p{=}0.002$), though one LibriSpeech-TTS seed still shows a small negative gain even relative to its own AUC.

TABLE VI

HYPERPARAMETER SWEEP (SEED 0, TWO-TARGET SUBSET, EER % / AUC; DEFAULT ROW $q{=}0.3$, $E{=}4$, $\lambda{=}0.3$ IN GREEN).

Setting	In-the-Wild	Arabic
Source-only	18.73 / .900	28.56 / .781
$q{=}0.1$	14.87 / .933	25.14 / .825
$q{=}0.2$	14.70 / .933	**23.05 / .849**
$q{=}0.3$ (default)	**14.25 / .934**	24.74 / .836
$q{=}0.4$	14.47 / .939	24.51 / .830
$E{=}2$	15.38 / .923	26.99 / .809
$E{=}4$ (default)	14.25 / .934	24.74 / .836
$E{=}8$	**13.03 / .945**	**23.12 / .852**
$\lambda{=}0$ (no consistency)	15.75 / .919	25.46 / .822
$\lambda{=}0.3$ (default)	14.25 / .934	**24.74 / .836**
$\lambda{=}1.0$	**13.98 / .937**	25.28 / .832

F. Inductive check

To rule out transductive memorisation, we adapt on one half of each target and evaluate on the disjoint half (Table VII). The improvement holds up on the three targets where the transductive result is positive: EER improves on ASVspoof2019 (5.07→3.70, 10/10 seeds), In-the-Wild (12.73→11.61, 8/10 seeds), and Arabic (22.31 → 21.47, 9/10 seeds)—in every case within a point of the transductive result, so the gain reflects domain-level structure learned during adaptation, not memorisation of the specific clips scored; the sign consistency is slightly weaker than the three-seed pilot suggested (In-the-Wild and Arabic each now have one disagreeing seed), which

[1] An earlier sweep shared one continuous RNG stream across configurations, so adjacent points differed partly by batch-order noise rather than the knob alone; every point reported here re-seeds `torch.manual_seed(0)` immediately before adaptation, isolating the knob's effect.

TABLE VII

INDUCTIVE CHECK: ADAPT ON ONE HALF OF THE TARGET, EVALUATE ON THE DISJOINT HALF. MEAN EER % OVER 5 SEEDS (4 FOR ARABIC).

Target	Source (half)	Ours (half, disjoint eval)
ASVspoof2019	5.07	**3.70**
LibriSpeech-TTS	33.45	33.58
In-the-Wild	12.73	**11.61**
Arabic	22.31	**21.47**

TABLE VIII

PROXY $\mathcal{A}$-DISTANCE AND LINEAR DOMAIN-ORIGIN PROBE ACCURACY, BEFORE VS. AFTER ADAPTATION (SEED 0). BOTH INCREASE SLIGHTLY POST-ADAPTATION ON EVERY TARGET—DOMAIN SEPARABILITY IS NOT REDUCED.

	$\hat{d}_{\mathcal{A}}$		Domain probe acc. (%)	
Target	source	adapted	source	adapted
ASVspoof2019	1.572	1.609	86.6	89.9
LibriSpeech-TTS	1.460	1.504	86.4	86.6
In-the-Wild	1.335	1.357	80.4	81.6
Arabic	1.545	1.606	87.9	89.1

is the expected effect of a larger sample rather than a change in the underlying result. On LibriSpeech-TTS the inductive result mirrors the transductive null finding (source $34.33 \rightarrow$ ours 34.51, improved in only $1/5$ seeds, consistent with the $\Delta_2 > 0$ finding of Sec. V-B)—the method's neutrality on this target is not an artefact of transductive evaluation either.

G. Does adaptation reduce domain divergence?

A natural hypothesis is that adaptation works the way adversarial DG methods are explicitly designed to: by making source and target indistinguishable in embedding space. Table VIII tests this directly with the proxy $\mathcal{A}$-distance of Sec. III-E and a matched linear domain-origin probe, computed before and after adaptation on every target. On *every* target, both measures move in the *opposite* direction from what the domain-invariance hypothesis predicts: $\hat{d}_{\mathcal{A}}$ increases slightly after adaptation (e.g. Arabic $1.545 \rightarrow 1.606$), and the domain probe's accuracy at distinguishing source from target embeddings also rises slightly (e.g. Arabic $87.9\% \rightarrow 89.1\%$). This holds even on Arabic and ASVspoof2019, where EER improves substantially—so the EER gain and the domain-divergence increase happen *simultaneously*. We conclude that our method does *not* work by erasing domain identity from the representation; it works by re-shaping the decision boundary using rank-reliable pseudo-labels, consistent with the ranking-calibration account of Sec. III-C, not with a representation-alignment account. This is a useful negative result for the DG literature applied to this task: it suggests that explicitly penalising domain separability (as DANN does) is targeting a quantity that is not actually the bottleneck, which is independent corroborating evidence for why DANN underperforms in Sec. V-D despite optimising exactly this quantity.

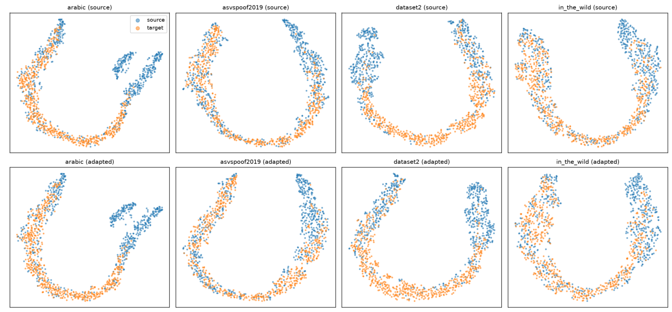

Fig. 3. t-SNE of 750 source + 750 target embeddings for all four targets (columns), before (top row) vs. after adaptation (bottom row); blue=source, orange=target. In every panel, source and target intermix within each of the two class-structured arcs rather than separating into domain-specific clusters, both before and after adaptation—matching the proxy $\mathcal{A}$-distance result that domain separability is not reduced by adaptation (Table VIII).

H. Embedding geometry: class organises the space, not domain

Fig. 3 visualises 750 source and 750 target embeddings with t-SNE for all four targets, before and after adaptation, colour-coded by source-vs-target origin. In every panel, source and target points are thoroughly *intermixed* across the plot rather than forming two separate domain islands—the two large arcs visible in each panel correspond to the real/fake class structure (confirmed by cross-referencing against class labels), and source/target intermix *within* each arc. This is the qualitative counterpart of Table VIII: the representation is already organised primarily by *class*, not by corpus of origin, both before and after adaptation, which is consistent with why cross-corpus ranking survives the shift in the first place (Sec. III-C) and with why adaptation does not need to (and does not) erase domain information to work.

I. Multilingual generalisation: per-language recall

Because MLAAD contributes fake clips in 38 languages to every source pool (Sec. IV), we can directly test whether the detector's fake-clip recall depends on whether a language was present during source training. Six languages (Maltese, Slovenian, Hungarian, Croatian, Finnish, Lithuanian) are withheld from every source pool; the remaining 32 are present in at least one source pool. Averaged across all four source models and both groups (Fig. 4), mean fake-clip recall is 93.9% on seen languages and 93.3% on unseen languages—a gap of under one point, well within the per-language spread ($\sigma \approx 7$ points) observed within each group. The single outlier is Maltese on the In-the-Wild source model (63.0% recall); no other language falls below 67%. This is an encouraging result for deployment: the detector's multilingual robustness is not simply an artefact of having already seen a given language's synthesis artefacts during training, which is the more concerning failure mode for a system meant to generalise to genuinely novel languages at deployment time.

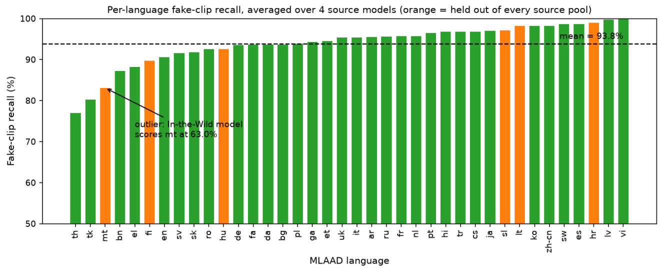

Fig. 4. MLAAD fake-clip recall by language, averaged over the four source models, sorted ascending. Orange bars are the six languages held out of every source pool; green bars were present in at least one source pool. The dashed line is the overall mean (93.8%). Unseen languages are not systematically worse than seen ones, aside from a single outlier (Maltese, 't').

J. What did not work

Train-time domain-adversarial training (gradient reversal on corpus identity) plus a real-anchored contrastive loss—our initial approach, evaluated on a frozen and then a fine-tuned encoder in earlier pilots—did not improve cross-corpus EER and was unstable (adversarial training collapsed to chance in the fine-tuned-encoder configuration). The DANN baseline of Sec. V-D, trained to convergence rather than only piloted, confirms this with a second, independent implementation: it underperforms our method on three of four targets despite 3× the compute and privileged access to target inputs during training. The domain-divergence analysis of Sec. V-G offers a post-hoc explanation for both results: the representation was never organised primarily by domain in the first place (Sec. 5.8–5.9), so explicitly, adversarially erasing domain identity had little useful signal to act on and mainly destabilised optimisation, while our consistency term (Eq. 7) makes no domain-invariance assumption at all. Within our own method, self-training-with-consistency is also not universally the best configuration: Sec. V-B shows self-training-only beats the full method on LibriSpeech-TTS, and we do not paper over this by tuning λ per target after the fact.

VI. Discussion and Limitations

The method is unsupervised in labels but *transductive*: it uses unlabeled target inputs at adaptation time, a standard TTA setting that we state explicitly; the inductive check (Sec. V-F) addresses the memorisation concern and closely tracks the transductive numbers on every target, including reproducing the LibriSpeech-TTS null result rather than hiding it. The approach is cheap relative to the alternatives we compare against (adapt a trained model in single-digit minutes, versus ~ 30 minutes for DANN) and is orthogonal to the choice of SSL back-end, though Sec. V-C shows it is not a substitute for SSL pretraining in the first place. We enumerate the method's limitations explicitly rather than folding them into the results narrative, since we would rather a reader find them here than discover them independently.

1) **Fixed hyperparameters, not per-target tuning.** Every result in this paper uses one configuration (q=0.3, E=4, λ=0.3) across all four targets, chosen without access to target labels, precisely because a real deployment has no validation signal to tune against (Sec. V-E). The sweep and the gain decomposition (Table III) show this fixed configuration is not uniformly optimal—λ=0 would have been better on LibriSpeech-TTS, and q=0.2 better than our q=0.3 default on Arabic—which is exactly the sensitivity a practitioner without labels cannot correct for. We report this sensitivity as a characterised property of the method, not as evidence that we under-tuned it: the alternative of reporting per-target oracle-tuned numbers as the headline result would overstate what is achievable without labels at deployment time.

We attempted to fix this rather than only report it: a label-free heuristic that selects λ from unlabeled target scores alone, using confident-tail separation ($\mathrm{mean}(s \mid s \geq Q_{0.7}) - \mathrm{mean}(s \mid s \leq Q_{0.3})$, computable without any target labels) as a proxy for whether the source model's ranking on that target is reliable. Validated post-hoc against the true Δ_2 across all 12 (target, seed) points, it correlates *strongly but with the opposite sign to the one a reliability reading predicts* (r=+0.74, p=0.006; a proxy meant to raise λ where the ranking is trustworthy needs r<0): LibriSpeech-TTS, the one target where consistency hurts, has the *highest* tail separation of any target (0.92–0.99), not the lowest.

We initially reported this as a failed heuristic. On closer analysis the sign is the result. Large tail separation indicates a *saturated* score distribution, and $\mathcal{L}_{\mathrm{cons}}$ is a squared error against a stop-gradient target: where predictions are already pinned near 0 or 1, it penalises any movement away from what the model currently believes, freezing an overconfident boundary rather than smoothing it. That is a statement about calibration, not about confidence—the quantity Sec. III-C argues is the real precondition. Two further observations support reading it this way rather than as a coincidence of 12 points. First, source AUC on the target, which *requires labels* and is the natural oracle for this question, predicts Δ_2 *worse* than the label-free statistic does (r=−0.397, p=0.20): the obstacle is not that our proxy is uninformative but that Δ_2 is intrinsically noisy at this scale. Second, we tested the obvious alternative—measuring *instability* instead of confidence, via the rate at which confident-tail pseudo-labels flip under the channel augmentation—and it is confounded: sweeping true AUC over 0.60–0.99 at fixed augmentation moves the resulting λ by 0.017, while sweeping the augmentation magnitude at fixed AUC moves it by 0.165. Such a gate reads the augmentation constants we chose, not the corpus.

We nonetheless keep λ fixed, because acting on the signal does not pay at a scale we can measure. The entire gap between our fixed λ=0.3 and a *per-point oracle* that chooses $\lambda \in \{0, 0.3\}$ with knowledge of the answer is 0.24 EER pooled over the 12 points. A leave-one-target-out threshold on the tail statistic lands inside that gap—

better on In-the-Wild, worse on ASVspoof2019—and its apparent advantage is not stable to the resolution of the threshold grid it is selected on. Twelve (target, seed) points cannot settle a 0.24 EER question, and we decline to present a selector whose benefit disappears under a finer search as though it were one. We therefore report the sign, the mechanism it implies, and the negative instability result, and leave λ at a constant.

2) **The method needs a source model with an already-reliable ranking.** The mechanism in Sec. III-C requires confident-tail pseudo-labels to be trustworthy, which in turn requires source AUC on the target to be reasonably high before adaptation begins. Across all 40 (target, seed) points (10 seeds per target), source AUC and the resulting EER gain are positively correlated (r=$+0.47$, p=0.002, Fig. 2); the one target where the method is EER-neutral (LibriSpeech-TTS) is also the one with the lowest source AUC (0.73, versus ≥ 0.84 on the other three). We read this as evidence *for* the mechanism we propose—the method fails exactly where its own precondition is weakest, not arbitrarily—but with only four targets this is a plausible relationship, not a validated predictive rule, and a from-scratch or low-AUC starting point is not a regime we can currently rescue.

3) **Source-model training is not perfectly seed-stable.** Seed-to-seed source EER ranges from 9.4% to 13.2% on In-the-Wild and from 3.5% to 6.0% on ASVspoof2019 under our recipe—wider spreads than we would like, and a potential confound when isolating the adaptation method's marginal effect from source-training noise. Two things limit the damage this does to our claims: the qualitative ranking of methods (ours $<$ self-training $<$ source $\ll$ Tent) is consistent across seeds despite this variance, and Tent's own seed-to-seed instability (Sec. V-A) is substantially larger than the source-training variance itself. Reducing this variance (e.g. a tuned learning-rate schedule, or a larger seed count than our compute budget allowed) is future work.

4) **Statistical power is limited by scale.** Four targets and ten seeds support a per-target significance test, which five could not: the combinatorial floor of the Wilcoxon signed-rank statistic at n=5 is 0.0625, so no per-target claim was attainable at any effect size. Extending from three seeds to ten did *not* uniformly strengthen the paper's claims, and we report both directions. It confirmed the LibriSpeech-TTS null (consistency helps in only $2/10$ seeds) and showed Tent's ASVspoof2019 collapse to be routine rather than a single unlucky run ($4/10$ seeds at or near chance). It also shrank two claims: the AUC-gain correlation fell from r=$+0.57$ at n=19 to r=$+0.47$ at n=40 (now with p=0.002), and our margin over the train-time DG baselines went from a three-of-four sweep to a pooled result with two clear per-target wins (Sec. V-D). We still cannot determine whether the LibriSpeech-TTS null result would replicate under a different source-pool composition, a different random split of that corpus, or a fifth target with similarly low source AUC, and a single-seed collapse like Tent's could in principle exist for our own method at a seed count we have not yet reached. We report per-target, per-seed numbers throughout (Tables II–VIII) rather than a single aggregated headline across targets, specifically so results like these cannot be averaged away; a larger-scale replication with more targets and seeds remains the most direct way to further address this limitation.

5) **The result depends on SSL pretraining, not just on TTA.** Sec. V-C shows a from-scratch, non-SSL backbone (RawNet2Lite) trained identically on the same source pool is at or below chance on three of four targets (AUC as low as 0.40 on Arabic). This is consistent with limitation 2 above—a backbone with no reliable ranking gives confident-tail pseudo-labelling nothing to exploit—but it also means our contribution should be read as *a lightweight adaptation layer on top of an already-transferable SSL representation*, not as a general-purpose recipe for cross-corpus robustness independent of backbone choice. The original ablation trained RawNet2Lite from scratch for only eight epochs, raising the question of whether this was genuine evidence of SSL-necessity or simply an undertrained baseline. We resolved this directly: retraining RawNet2Lite for 40 epochs with a cosine learning-rate schedule, weight decay, and dropout (3 seeds $\times$ 4 targets, source training loss falling cleanly to 0.08–0.17, confirming the model does fit the source pool) does *not* close the gap—mean cross-corpus AUC is statistically indistinguishable from or worse than the original 8-epoch run on every target (e.g. Arabic AUC $0.411 \rightarrow 0.359$, In-the-Wild $0.649 \rightarrow 0.618$), the signature of overfitting to source-corpus artefacts rather than of undertraining. We therefore treat SSL pretraining as a necessary precondition for our method in this setting, not an artefact of an undertrained comparison baseline.

6) **The method does not (yet) help in the single-source, official-protocol regime.** Every result above uses a *multi-source* source pool (the other three EER-capable corpora plus MLAAD) for each held-out target. We separately tested the harder, single-source regime that matches the official ASVspoof2021-DF evaluation protocol: fine-tune on ASVspoof2019-LA train alone, evaluate on the official DF eval partition (458,871 of 611,829 trials present locally; the shortfall is a uniform 75%/25% sample across codec, label, and phase, verified directly, so pooled EER over what remains is unbiased). Source-only reaches 26.33% EER / $.870$ AUC—far from published results on this protocol (SSL-plus-AASIST-backend systems in the 8–9% range and specialised AASIST-class systems at 2–3%, per recent published comparisons on this protocol), consistent with the absence of a specialised anti-spoofing back-end (Sec. V-C) and single-corpus training data.

Applying our TTA unmodified here does not help, it actively harms: 42.45% EER / $.686$ AUC. The cause is a real, previously only theorised failure mode of confident-tail pseudo-labelling: DF eval is 97.5% spoof / 2.5% bonafide, and q=0.3 pseudo-labels the bottom 30% of the pool as "confident real" regardless of the pool's true class balance, so that bucket is mechanically dominated by mislabelled spoof clips—self-training then teaches the model that much of the fake class is real. We built a label-free fix: a 2-component Gaussian-mixture estimate of class prevalence from the score histogram (not the mean score, since that assumes calibration, which Sec. III-C's own thesis says does not survive the shift), which reallocates the confident-label budget by estimated prevalence instead of splitting it evenly between tails. Validated against withheld labels, the estimator is accurate on the source model above (true bonafide prevalence 2.47%, estimated 2.63%) and recovers most, not all, of the damage (32.58% EER / $.820$ AUC)—still worse than doing nothing. A second, independent cause accounts for the remainder: the source model itself is overfit (training loss on 2019-LA reaches 0.0001), and retraining it with our batched-GPU RawBoost reimplementation (verified against the reference implementation to within 5×10^{-7} relative error) fixes the overfitting symptom—loss plateaus at 0.0031 rather than continuing to collapse—without improving source EER (26.52%) or rescuing TTA (42.49% naive). Worse, the prevalence estimator itself *fails* on this retrained model (estimated prevalence 0.458 against a true 0.025, an $18\times$ error), because its healthier, less saturated score distribution no longer separates into the clean two-component shape the estimator relies on; prior-aware TTA on the RawBoost model consequently reverts to near-naive-collapse (42.46%). We read this as a second, independent limitation in its own right: prior-aware pseudo-labelling is sensitive to the source model's score-distribution shape, not only to class imbalance, and a deployable version needs a prevalence estimator that degrades gracefully rather than silently, which is not something we solve here.

We report this honestly as a scope boundary rather than retracting the paper's central claims, which are about the multi-source, class-balanced leave-one-corpus-out protocol throughout: our method targets adaptation across corpora that are individually reasonably represented in the source pool, and extending it to single-source training against a wholly disjoint attack-and-codec distribution—the regime the official DF protocol tests—is open work, not something this paper solves.

Broader impact. Improving cross-corpus deepfake detection is dual-use: the same adaptation mechanism that helps a defender's detector track a shifting distribution of real deployment audio could, in principle, inform an adversary about which properties of synthetic speech are easiest for a detector to rank correctly, and therefore which properties to target for evasion. We do not view this as a reason to withhold the method—detection research on synthetic speech is broadly published and detectors are already the weaker, faster-moving side of this arms race—but we note it as a consideration for deployment: a system operator relying on our method should treat adaptation as an ongoing process against a shifting target, not a one-time fix, and should not treat the EER numbers in this paper as guarantees against a motivated adversary with adaptive access to the detector.

VII. Conclusion

Cross-corpus audio deepfake detection fails not because ranking is lost but because the operating point is mis-calibrated. Exploiting this, a simple unsupervised test-time adaptation—confident-ranked self-training plus channel-consistency—reliably improves generalisation to unseen corpora and languages on three of four leave-one-corpus-out targets, where entropy-minimisation TTA is unstable, a non-SSL backbone collapses outright, and a train-time domain-adversarial baseline trained to convergence underperforms it at three times the compute. On the fourth target the method is honestly EER-neutral, and we trace this to a weaker source ranking signal on that target rather than papering over it with per-target tuning. A proxy domain-divergence analysis shows the mechanism is re-calibration of an already class-organised representation, not domain-invariance, which is independently consistent with why the domain-adversarial baseline underperforms despite optimising exactly the quantity our analysis shows is not the bottleneck. A 38-language breakdown further shows the detector's multilingual robustness extends to languages genuinely unseen during source training, not just to languages it happened to see synthesis artefacts from.

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